

Research Article

Study on the Multitarget Synergistic Effects of Kai-Xin-San against Alzheimer's Disease Based on Systems Biology

Sirui Guo,^{1,2} Jiahong Wang^{ID},³ Yunjia Wang,¹ Ying Zhang,¹ Kaishun Bi,¹ Zhou Zhang^{ID},³ and Qing Li^{ID}¹

¹School of Pharmacy, Shenyang Pharmaceutical University, Shenyang, 110016 Liaoning, China

²Department of Pharmacy, Beijing Hospital, National Center of Gerontology, Institute of Geriatric Medicine, Chinese Academy of Medical Sciences, Beijing 100730, China

³School of Life Sciences and Biopharmaceutics, Shenyang Pharmaceutical University, Shenyang, 110016 Liaoning, China

Correspondence should be addressed to Zhou Zhang; zzhouzhang@163.com and Qing Li; lqyxm@hotmail.com

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Kai-Xin-San (KXS), a classical Chinese traditional prescription, was widely applied in the treatment of Alzheimer's disease (AD), while its functional mechanisms still remain unclear. By using systems biology approaches at animal, cellular, and molecular levels, the improvement of KXS on cognitive impairment was achieved by inhibiting abnormal acetylcholinesterase. The function on the nerve skeleton was performed by regulating the Tau phosphorylation pathway. Its antioxidant, anti-inflammatory, and antiapoptotic effects by modulating the aberrant upregulation of ROS, proinflammatory factors, and apoptosis-related proteins in the brain were studied to reveal the synergistic therapeutic efficacy of KXS. Then, formula dismantling *in vitro* indicated that ginseng was the principal herb, whereas three other herbs served adjuvant roles to achieve the best effect. After that, the *in vivo* analysis of components into plasma and brain of AD rats showed that 8 of 23 components in blood and 4 of 10 components in brain were from ginseng, respectively, further verifying the principal status of ginseng and the synergistic effects of the formula. Thus, the anti-AD effects of KXS were achieved by multitargets and multichannels. The systems biology approaches presented here provide a novel way in traditional herbal medicine research.

1. Introduction

Herbal medicine formulae with unique advantages have been widely used for thousands of years based on the guidance of traditional Chinese medicine (TCM) theory, a combined therapeutic system in fighting disease [1]. Generally, a formula is composed of several Chinese herbal medicines. The herbs with the strongest pharmacological action have a principal role, while others play an assisting role to strengthen the effect or minimize the adverse effects [2]. It is well known that Chinese herbal formulae have holistic treatment and synergy strategies; there are many components in TCM [3]. However, how these herbs can be combined in one formula and create a powerful therapeutic effect still remains confusing. Therefore, it is important to study the synergistic effects

and the underlying mechanisms of Chinese herbal medicine formulae.

Alzheimer's disease (AD), a widespread neurodegenerative disease, is characterized by the deposition of intercellular amyloid polypeptides and intracellular neurofibrillary tangles (NFTs), which are mainly composed of hyperphosphorylated Tau [4]. AD is also triggered by other complicated pathogenic factors such as oxidative stress, neuroinflammation, and neuronal apoptosis, and there is a vicious circle among them in many patients [5]. It was reported that there are more than 50 million cases of dementia worldwide in 2018, signifying that one new case occurs every 3 seconds [6]. The increasing prevalence of dementia not only caused pain to the patients but also imposed a heavy burden on the affected families and the whole society [7, 8]. The annual total cost of

dementia was \$167.74 billion in 2015, and it is predicted to reach \$507.49 billion in 2030 and \$1.89 trillion in 2050 [9, 10]. Currently, all FDA-approved drugs were developed for a specific single target, such as cholinesterase inhibitors (donepezil, rivastigmine, and galantamine) and N-methyl-D-aspartate receptor agonist (memantine) [11]. However, their therapeutic effects were below our expectations because their actions on a single target could not match multiple pathogenetic factors mainly including abnormal β -amyloid and Tau [12–14]. Interestingly, with the multi-target and multicomponent treatment characteristics, TCM had unique advantages in dealing with AD [15, 16]. It has been proven that *Panax ginseng*, one of the main representatives of TCM, was effective in treating AD, and the key mechanism has been revealed to be related to the PI3K/Akt signaling pathway [17]. A series of formula studies on the *Panax ginseng*/Polygala tenuifolia combination yielded varying improvements in the course of AD and the most representative formula was Kai-Xin-San (KXS), initially recorded in *Bei Ji Qian Jin Yao Fang* in Tang dynasty (AD 652) [18, 19]. KXS consists of four single herbs which are used in treating AD frequently, namely, *Polygala tenuifolia* Willd. (PR), *Panax ginseng* C. A. Mey. (GR), *Poria cocos* (Schw.) Wolf (PO), and *Acorus tatarinowii* Schott (AT) [20]. Recent researches of KXS were mainly focused on component identification or mechanism exploration, and most pathological studies were related to single pathway neurotransmitter regulation, which could not completely explain the therapeutic mechanism of KXS and the synergistic effects among the four single herbs [21, 22]. Therefore, there is an urgent need to explain the multidimensional beneficial effects and synergism of KXS with systems biology methods.

In order to explore the principle of compatibility in TCM, the treatment of AD with KXS was applied as the working model. $A\beta_{25-35}$ - and D-gal-induced rats and $A\beta_{25-35}$ -induced PC12 cells were applied to establish AD models [23, 24]. Hence, we explained the therapeutic effects of KXS at the molecular, cellular, and animal levels and expounded the biological mechanism and synergism in fighting AD multidimensionally, which could provide research strategies in studying traditional formula.

2. Materials and Methods

2.1. Preparation of KXS. The decoction pieces of GR, PR, PO, and AT were purchased from Tongrentang Pharmacy. KXS was prepared using GR, PR, PO, and AT at a weight ratio of 3:2:3:2, then extracted by double-distilled water (1:10 w/v) for 1 h, and repeated a second time. The mixed extracted solutions were concentrated to 1 mg/mL by a rotary evaporator. The extraction method of four single herbs was consistent with the formula, and the concentrations were calculated based on the total dose.

2.2. Experimental Animals. Male Sprague-Dawley (SD) rats weighing 200–250 g (about 6 weeks old) were obtained from the Experimental Animal Center of Shenyang Pharmaceutical University. SD rats were kept in a controlled environment (temperature: $23 \pm 2^\circ\text{C}$; humidity: $50 \pm 10\%$) under a 12 h

light/12 h dark cycle with water and standard diet ad libitum. They were kept for 7 days to adapt to the new surroundings. Rats were randomly divided into eight groups ($n = 12$). Briefly, the AD rats were injected intraperitoneally with D-gal (50 mg/kg/mL) for 6 weeks and the control group was given the same volume of saline. Six groups of AD rats were administered intragastrically with KXS (10.0 g/kg), GR (3.0 g/kg), PR (2.0 g/kg), PO (3.0 g/kg), AT (2.0 g/kg), and Hup A (0.027 mg/kg) once a day from the third week until the end of the experiment [25]. At the end of the fourth week, the stereotaxic injection of $4\mu\text{g } A\beta_{25-35}$ ($1\mu\text{g}/\mu\text{L}$) was operated at the following coordinates: anteroposterior—3.5; mediolateral—+2.0; and dorsoventral—2.8 mm [26]. Two weeks later, the Morris water maze test was used for evaluation. All rats were sacrificed with intraperitoneal injection of chloral hydrate (350 mg/kg). The experimental operations were approved by the Guidelines for Animal Experimentation of the Shenyang Pharmaceutical University and the Animal Ethics Committee of the institution.

2.3. Morris Water Maze Test. At the seventh week, Morris water maze performance was applied to evaluate a rat's spatial learning and memory ability [27]. The rats were trained to find a hidden safety platform located in the same position (1 cm below the water surface, 10 cm in diameter), and complete training was conducted twice per day for 4 days. The training trial was finished if the rat stood on the safety platform and stayed for at least 10 s. Those rats that could not find the hidden platform within 90 s were guided to the platform and kept on the platform for 10 s. The probe test proceeded 24 h after the last training trial. The rats were tested to swim freely to find the removed platform within 90 s. The distance and time in the target quadrant as well as the times of crossing the platform were recorded to evaluate spatial memory. The drug administration and Morris water maze test were scheduled according to the experimental timelines (Figure 1) and followed the methods of Guo et al. 2019 [28].

2.4. Immunofluorescence and Immunohistochemistry. The expression of neuronal nuclei (NeuN) was determined with immunofluorescence. Brain tissue sections ($4\mu\text{m}$ thickness) were rehydrated with gradient alcohol, antigen repaired with citric acid buffer (pH 6.0), and then blocked with BSA for 30 min. The sections were incubated with NeuN antibody (Servicebio, 1:300) at 4°C overnight, followed by CY3-labeled secondary antibodies (Servicebio, 1:500) at room temperature for 60 min. For nucleus staining, the sections were incubated with DAPI fluorescent dye for 8 min and mounted with antifade mounting medium. Images of the immunofluorescence slices were scanned with Nikon Eclipse C1. The positive expression in the hippocampal CA1 region (3 sections per rat) was dealt with the Image-Pro Plus Software.

The expressions of phosphorylated Tau in hippocampus CA1 were operated with immunohistochemical analysis. Histologic sections ($4\mu\text{m}$ thickness) were antigen repaired with citric acid buffer (pH 6.0), incubated with 3% H_2O_2 for 25 min, and then blocked with 3% BSA for 30 min.

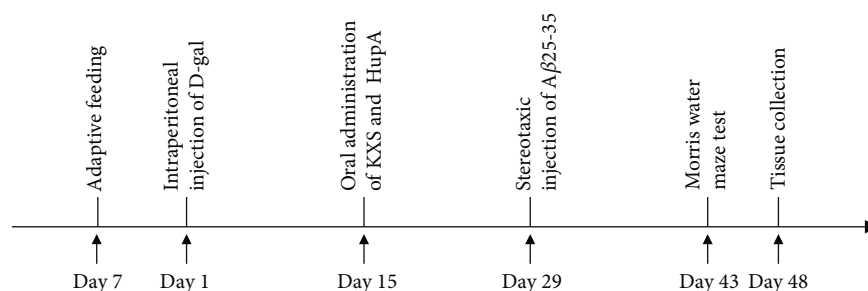


FIGURE 1: The schematic diagram of the AD rat experimental procedure.

Subsequently, the sections were incubated with monoclonal anti-Tau (phosphor S396) antibody (Abcam, 1:1000) at 4°C overnight, followed by streptavidin-horseradish peroxidase (Servicebio, G1211) for 60 min at room temperature. DAB color-substrate solution (Servicebio, G1211) was applied for coloring, and the images of the immunohistochemical sections were acquired by a CIC microscope (XSP-C204).

2.5. Western Blotting. The hippocampus was harvested and lysed with lysis buffer (Beyotime Biotechnology, China) containing 1% phosphatase inhibitors and protease inhibitor. The quantitative proteins (10 µg) were separated on 10% SDS-polyacrylamide gels and transferred to a PVDF membrane. The PVDF membrane was blocked with 5% skim milk powder in Tris-buffered saline containing 0.1% Tween 20 (TBS-T) for 2 h and then incubated in the primary antibodies diluted in blocking solution overnight at 4°C. The details of the primary antibodies are listed in Supplementary Table 1. Then, the PVDF membrane was washed with TBS-T and incubated in corresponding secondary antibodies at room temperature for 2 h. After that, the PVDF membrane was covered with ECL and photographed with the Chemiluminescent Imaging System (Tanon Science & Technology Co., China). β -Actin was calculated as a control for protein expression in every membrane.

2.6. Detection of Acetylcholinesterase (AChE) Activity, Proinflammatory Factors, and Reactive Oxygen Species (ROS) In Vivo. Brain tissue sections in the hippocampus were homogenized with 10 volumes (1:10, *w/v*) 0.1 mM phosphate-buffered solution (PBS) and centrifuged at 1,000 g for 10 min at 4°C. The supernatants were used for measuring the activity of acetylcholinesterase according to complimentary instructions (Nanjing Jiancheng, China). The absorbance was evaluated with a microplate reader at 412 nm. The AChE activity was calculated as follows: activity (U/mgprot) = $[\text{OD}_{\text{sample}} - \text{OD}_{\text{control}}] / [\text{OD}_{\text{standard}} - \text{OD}_{\text{blank}}] \times \text{standard concentration} / \text{protein concentration}_{\text{sample}}$. The supernatants were collected to also measure the proinflammatory factor (TNF- α and IL-1 β) levels in the hippocampus using an enzyme-linked immunosorbent assay (Elisa) kit (Boster Biological Technology Co. Ltd., China) according to the manufacturer's detailed instructions. The absorbance was evaluated with a microplate reader (TECAN,

Switzerland) at 450 nm. The level of ROS was measured with the ROS assay kit (Nanjing Jiancheng, China). According to the description of a previous study [29], the fluorescence in fresh hippocampal tissue was evaluated with the excitation at 500 nm and the emission at 525 nm.

2.7. Cell Culture, Drug Treatment, Cell Viability, Proinflammatory Factors, and ROS In Vitro. PC12 cells were obtained from the Boster Biological Technology Co. Ltd. (Wuhan, China) and cultured in H-DMEM supplemented with 1% L-glutamine, 1% penicillin/streptomycin, and 10% FBS (Gibco). Cells were incubated at 37°C in moist air containing 5% CO₂. In brief, PC12 cells were seeded in culture plates (1×10^5 cells/mL) for 24 h. Then, the cells were treated with KXS (1 mg/mL), GR (0.3 mg/mL), PR (0.2 mg/mL), PO (0.3 mg/mL), and AT (0.2 mg/mL), respectively, for 24 h. Subsequently, A β_{25-35} was added to the medium and incubated for an additional 24 h. Cell viability was measured by the 3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyltetrazolium bromide (MTT) assay. After different drug treatments, the treated PC12 was incubated with 100 µL MTT (1 mg/mL, Sigma-Aldrich) in a culture solution for 4 h and followed by the addition of 150 µL dimethyl sulfoxide (DMSO). The absorbance was detected at 490 nm wavelength by a microplate reader. Meanwhile, the supernatant after different drug treatments was collected to determine the level of IL-1 β and TNF- α . The concentrations of IL-1 β and TNF- α were measured according to the standard protocol from the manufacturer of Elisa assays (Boster Biological Technology Co. Ltd., China). The absorbance was evaluated with the microplate reader at 450 nm. Intracellular ROS was measured with the ROS assay kit (Beyotime Biotechnology, China). 2,7-Dichlorofluorescein diacetate (DCFH-DA) was a sensitive intracellular reactive oxygen detection probe widely used. Then, treated PC12 cells were loaded with 5 µmol/L DCFH-DA for 30 min at 37°C and washed with PBS for three times. According to a previous study [30], the fluorescence in PC12 cells was evaluated with the excitation at 500 nm and the emission at 525 nm.

2.8. Tubulin-Tracker Red Analysis. A β_{25-35} -induced cytoskeletal change of the PC12 cell was evaluated by a laser scanning confocal microscope. In brief, cells were prepared on 20 mm diameter coverslips in 24-well cell culture plates at 1×10^4 cells/well. After the drug treatment, the cytoskeletal changes of the PC12 cell's nucleus and microtubules were

evaluated by fluorescent visualization with DAPI (a blue fluorescent dye) and Tubulin-Tracker Red (a red fluorescent probe for microtubules), respectively. After the drug treatment, cells were washed twice in PBS and fixed with 4% paraformaldehyde for 20 min. Subsequently, the fixed cells were permeabilized twice with 0.1% (v/v) Triton X-100 in PBS for 5 min, incubated with Tubulin-Tracker Red (1:100) for 1 h, and then washed three times in 0.1% (v/v) Triton X-100. For nucleus staining, the fixed cells were incubated with DAPI fluorescent dye for 8 min and then washed and dried before use. The fluorescently labeled cells were examined by a C2 plus laser scanning confocal microscope (Nikon, Japan).

2.9. Measurement of the Characteristic Constituents of KXS In Vivo. The plasma samples collected from six rats after the Morris water maze test were mixed to eliminate individual variation. Aliquots of 1 mL mixed plasma samples were extracted with 3 mL methanol by vortex mixing for 3 min. After centrifugation at 12000 rpm and 4°C for 5 min, the supernatants were transferred and evaporated to dryness at 35°C. Then, the residue was reconstituted in 100 μ L methanol for HPLC-TOF-MS/MS analysis.

The HPLC system (Agilent 1260) and quadrupole time-of-flight mass spectrometry constituted the HPLC-QTOF-MS system. Chromatographic separation was achieved with a Phenomenex Kinetex XB-C18 Column (100 mm $\times$ 4.6 mm, 2.6 μ m) at 30°C. Mobile phases were the water (a) and methanol (b). A mobile phase was achieved with the following gradient elutions: 95%A $\rightarrow$ 75%A at 0–10.00 min; 75%A $\rightarrow$ 60%A at 10.00–25.00 min; 60%A $\rightarrow$ 25%A at 25.00–35.00 min; 25%A $\rightarrow$ 10%A at 35.00–41.00 min; 10%A $\rightarrow$ 10%A at 41.00–45.00 min; 10%A $\rightarrow$ 95%A at 45.00–50.00 min; and 95%A $\rightarrow$ 95%A at 50.00–55.00 min at a flow rate of 0.5 mL/min with a sample injection volume of 5 μ L. Mass spectrometric detection was operated by a Triple TOF 5600 (AB SCIEX, Foster City, CA) with both negative and positive ESI. The optimized parameters are listed as follows: ion spray voltage—5500 V (positive mode) and -4500 V (negative mode); source temperature—550°C; nebulizer gas as Gas 1—50 psi; heater gas as Gas 2—50 psi; curtain gas as Gas 3—30 psi; and declustering potential—80 V (positive mode)/-80 V (negative mode). Information-dependent acquisition was operated with the mass range of m/z 100–2000 to carry the MS/MS experiment. The collision energy of 60 $\pm$ 15 V with a collision gas of 15 psi was applied for information-dependent acquisition. The acquisition, instrument control, and analysis of data were dealt with Analyst TF 1.6 Software (AB SCIEX, USA).

2.10. Statistical Analysis. All statistical data were analyzed by Student's *t*-test and one-way ANOVA with post hoc Duncan's multiple range test (SPSS 19.0 software). The HPLC-QTOF-MS data acquisition was administered by the Analyst TF 1.6 Software (AB SCIEX, USA), and the data were further processed with the PeakView Software (version 2.2, AB SCIEX, USA). The statistical results were expressed

as mean $\pm$ SD, and statistical significances were classified as $P < 0.05$.

3. Result

3.1. Therapeutic Efficacies of KXS on Cognition Impairment of an AD Rat Model. In SD rat injected with D-gal and $A\beta_{25-35}$, a significant decrease of escape latency and swimming distance was observed in rats treated with KXS compared with other groups treated with vehicle control, positive control, or monotherapy of GR, PR, PO, and AT (Figure 2(a)). In the single-herb treatments, GR showed the most effective therapy. In the spatial probe, the rats treated with KXS showed a significant increase on the time and swimming distance in the target quadrant to find the removal platform (Figures 2(b) and 2(c)). In addition, KXS group rats showed more platform crossings than the other treatment groups (Figure 2(d)). Consistent with this, the expression of NeuN in the hippocampal CA1 regions of rats treated with KXS was higher than that in other treatment groups (Figures 2(e) and 2(f)). GR or PR alone also induced the expression of NeuN to some extent. Like Hup A, the AChE inhibitor, KXS could reduce the activity of AChE. Interestingly, GR or PR inhibited D-gal- and $A\beta_{25-35}$ -triggered augmentation of AChE, whereas KXS further intensified this effect (Figure 2(g)). Previous studies showed that the concentration of ACh was significantly reduced in the AD model with HPLC-MS/MS quantitative analysis [31]. It was suggested that KXS could ameliorate ACh loss (correlated with cognition impairment of AD) by suppressing the activity of AChE. Furthermore, the KXS protocol did not cause weight loss in AD rats (Figure 2(h)). These results suggested the KXS ameliorate cognition impairment without side effects on the AD rat model.

3.2. KXS Inhibited Oxidative Stress and Neuroinflammation-Induced Apoptosis In Vivo. By fluorescence analysis of the DCF fluorescence intensity in the hippocampus of rats induced by D-gal and $A\beta_{25-35}$, it was found that GR, but not PR, PO, or AT, induced an apparent decline of ROS levels, whereas KXS played the most important role compared with mono-GR, PG, PO, and AT (Figure 3(a)). It was tested if KXS treatment could act on proinflammatory factor expression by Elisa detection. Alone, GR had the greatest reduction in the concentration of TNF- α and IL-1 β in four monoherbs of KXS, whereas KXS further enhanced the anti-neuroinflammation effect (Figures 3(b) and 3(c)). Previous studies demonstrated that the accumulation of ROS and pro-inflammatory factors triggers the caspase cascade effect and causes apoptosis in the AD pathological process [32]. Bax, bcl-2, and cleaved-caspase-3 proteins were critical for the mitochondria apoptosis pathway and were abnormally expressed in AD. It was found that in the hippocampus of AD rats treated with KXS, bcl-2 at the protein level was downregulated, whereas Bax and cleaved-caspase-3 were upregulated (Figure 3(d)). Further analysis showed that a single GR or PR ameliorated the abnormal expression of these proteins, and the KXS formula enhanced the regulating effect. These results suggested that the KXS formula could

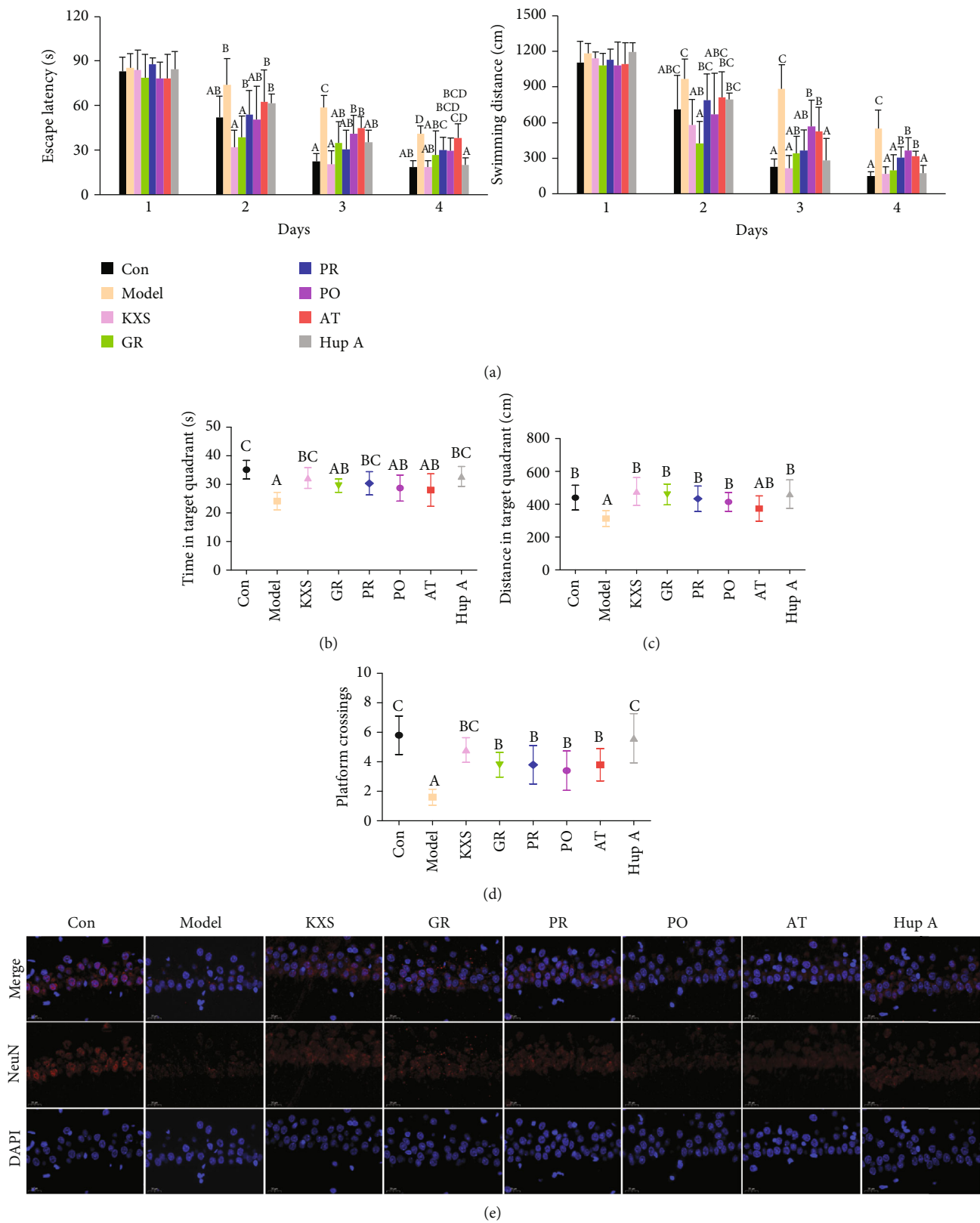


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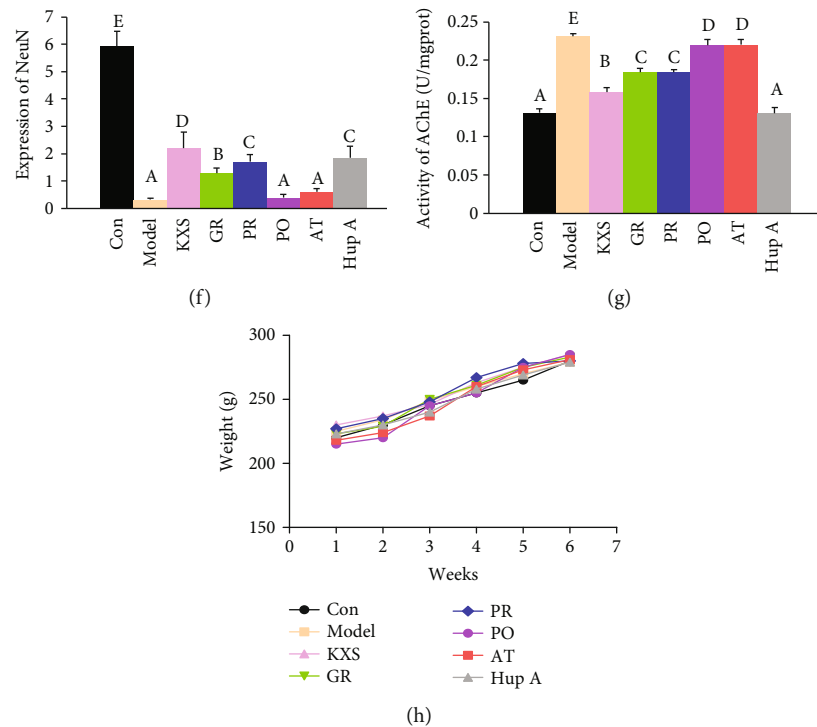


FIGURE 2: The ameliorative effect of KXS on AD rat memory and learning impairment. KXS significantly shortened the escape latency and swimming distances (a) of rats bearing cognition impairment compared with those treated with a monotreatment of GR, PR, PO, and AT. KXS prolonged the time (b), swimming distance (c), and platform crossings (d) to find the removal platform in the target quadrant. Treatment with KXS resulted in the high expression of NeuN (e and f) and the activity of AChE (g) in the hippocampus. KXS did not cause loss of body weight, suggesting that KXS might probably not cause severe toxicity (h). The column diagrams marked by different letters were significant to each other at the 0.05 level according to Duncan's multiple range test. Values were mean $\pm$ SE ($n = 6$), scale bar = 20 μ m.

inhibit oxidative stress and neuroinflammation-induced apoptosis; GR played the primary role in the formula, whereas PR, PO, and AT acted as adjuvant ingredients.

3.3. KXS Inhibited the Phosphorylation of Tau via PI3K/Akt Signaling Pathway In Vivo. Abnormal or hyperphosphorylated Tau was the principal pathogenetic factor in the pathogenesis of AD [33]. Previous studies showed that GSK-3 β was the major protein kinase to trigger the hyperphosphorylated Tau in the AD brain [34]. By western blot analysis of the proteins related to the Tau signaling pathway in the AD rat hippocampus, it was found that KXS could upregulate the expressions of p-PI3K, p-Akt, and p-GSK-3 β (S9), further inhibiting Tau hyperphosphorylation (Figures 4(a) and 4(b)). Among the four single herbs of KXS, GR played the principal role in the suppressive effect of hyperphosphorylated Tau (S199 and S396). Immunohistochemical staining with the anti-p-Tau (S396) antibody was applied to evaluate the expression of p-Tau in hippocampal CA1 regions. It was found that GR and PR could significantly downregulate p-Tau (S396), whereas the KXS formula further strengthened this effect (Figures 4(c) and 4(d)). These results suggested that KXS could inhibit the phosphorylation of Tau via the PI3K/Akt signaling pathway in AD rats and indicate that there could be a synergistic effect in the four single herbs of KXS.

3.4. KXS Causes Synergistic Effects on A β_{25-35} -Induced PC12 Cells In Vitro. To further validate the rationality of the four single herbs, *in vitro* researches were performed to test if PR and/or PO and/or AT combined with the administering of the GR herb exerted additive or synergic effects on PC12 cells. In PC12 (1×10^5 cells/mL) co-incubated with 20 μ mol/L A β_{25-35} , a statistically significant increase of activity was detected in cells treated with the formula KXS compared with PC12 cultured with model control or combined therapy of GR, PR, PO, and AT. Intriguingly, PR, PO, or AT improved GR-enhanced activity of A β_{25-35} -induced PC12 cells, whereas the KXS formula further promoted this effect (Figure 5(a)). Consistent with this, released proinflammatory factors (TNF- α and IL-1 β) in KXS-treated PC12 cells were lower than in other treatment groups (Figures 5(b) and 5(c)). Cells, upon KXS treatment, showed higher activity to inhibit oxidative stress detected by the ROS levels, and showed downregulated ROS levels compared with model control or combined therapy of GR, PR, PO, and AT (Figure 5(d)). Previous studies showed that Tau hyperphosphorylation could reduce microtubule stability and influence the expression of Tubulin in the AD model [35]. It was found that GR alone ameliorated a certain degree of the microtubule morphology. What's more, the stabilization effect on microtubules was further enhanced in A β_{25-35} -induced PC12 cells with the increase of compatible herbs

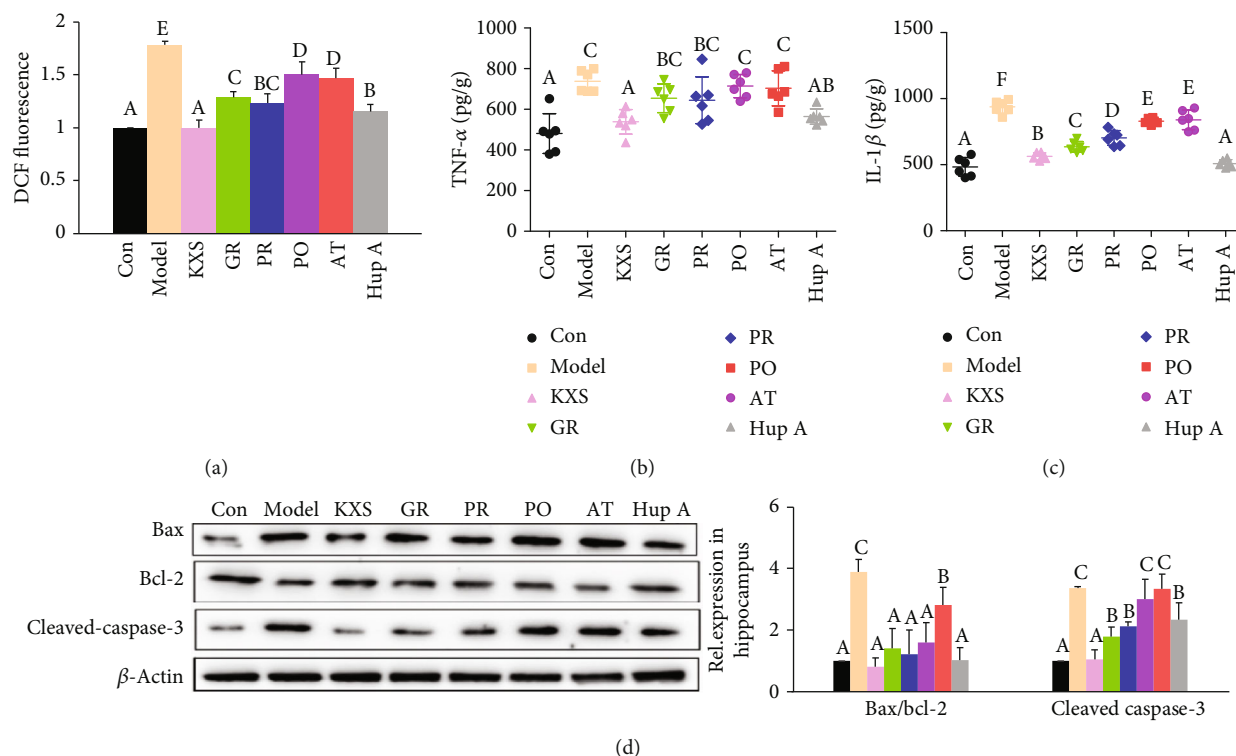


FIGURE 3: Effect of KXS on oxidative stress and neuroinflammation-induced apoptosis. KXS treatment relieved the accumulation of ROS in the rat hippocampus (a). Effect of KXS on representative proinflammatory factors TNF- α (b) and IL-1 β (c). Downregulation of apoptosis-related proteins Bax and bcl-2 by KXS treatment in the rat hippocampus resulted in a decrease of cleaved-caspase-3. Values were mean $\pm$ SE ($n = 6$). The column diagrams marked by different letters were significant to each other at the 0.05 level according to Duncan's multiple range test.

(Figures 5(e) and 5(f)). Consistent with *in vivo* results, these results suggested the synergic effects in the four herbs, and GR played the principal role in the formula, whereas PR, PO, and AT could serve as adjuvant herbs.

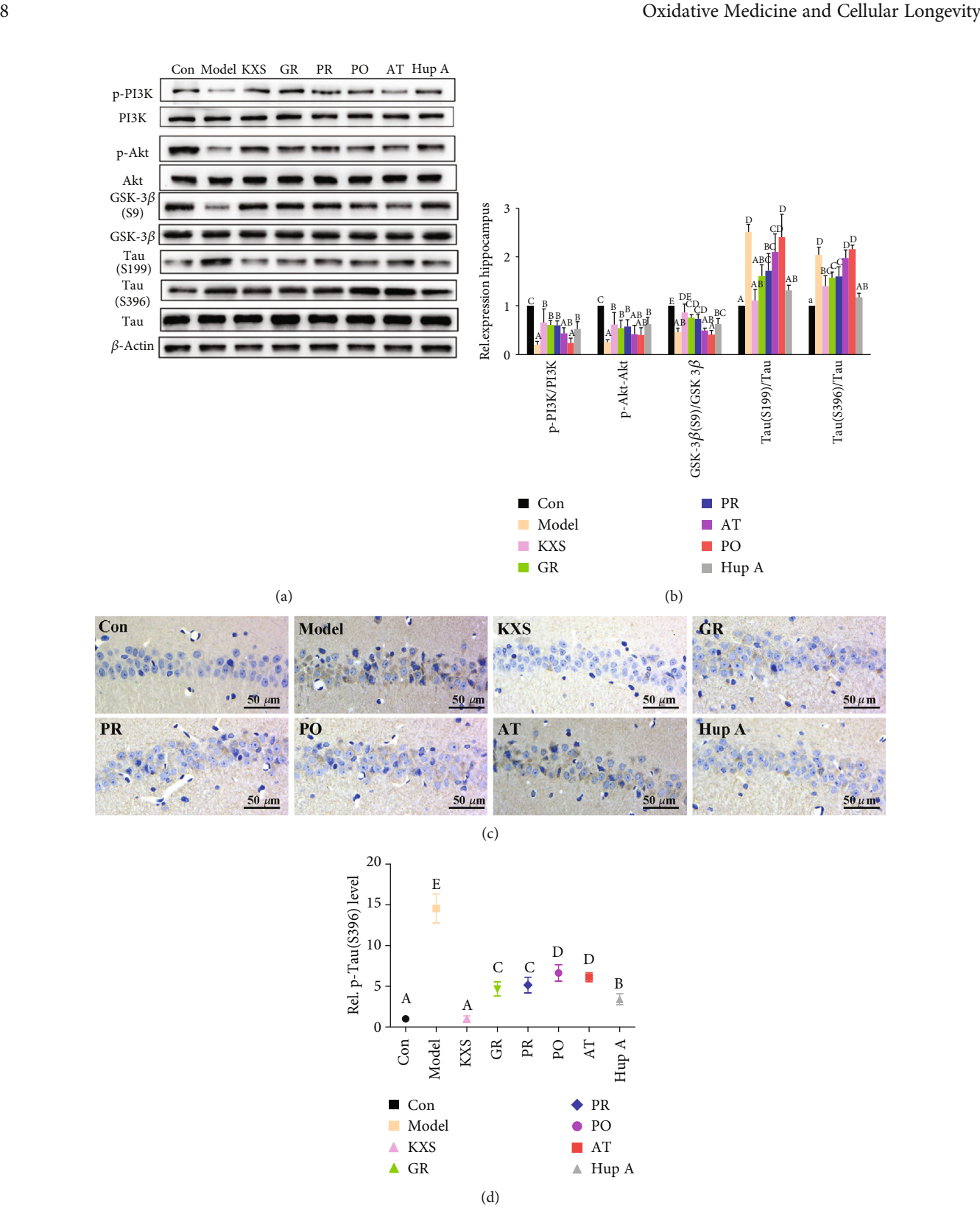
3.5. Analysis of the Basic Pharmacodynamic Material of KXS at the Small Molecular Level. After the optimization of chromatographic separation conditions and MS detection parameters, an appropriate HPLC-QTOF-MS method was developed to explore the ingredients of KXS in the AD rat model. The multiple product ion filtering (mPIF) technique based on multiple diagnostic product ions and the neutral loss filtering (NLF) technique was applied to identify compounds *in vivo* [21]. In the full-scan mass spectra, the compounds exhibited $[M+H]^+$ or $[M+Na]^+$ ions in positive ion mode and $[M-H]^-$ in negative ion mode. In total, 23 components were firstly found in AD rats' plasma (including 8 from GR, 1 from PR, 4 from PO, 6 from AT, and 4 that only existed in KXS) by analyzing blank and drug-containing plasma samples of AD rats. 10 constituents were firstly detected in AD rats' brain including 4 from GR and 6 only from KXS. The order of the isomers was preliminarily determined according to polarity. The compounds and their MS data are listed in Table 1 and Table 2. The peak area of components *in vivo* after oral administration of KXS in AD rats is presented in Supplementary Tables 1–2. The main effective composition *in vivo* was preliminarily elucidated, and the mass accuracy of all identified compounds was less than

5 ppm. Based on these, it was indicated that there were synergic effects among the four single herbs and KXS facilitated the transportation of GR into plasma and brain.

3.6. Correlation Analysis of Plasma Components and Pharmacological Effects of KXS. To explore the relationship between plasma components and pharmacological effects of KXS, the Pearson correlation analysis was performed with SPSS 19.0 software to rank four single herbs in an anti-AD effect [36, 37]. To evaluate the absolute value of the coefficient as the degree of correlation, we performed a semi-quantitative analysis of each prescription chromatogram (Table 3). When the Pearson correlation coefficients were greater than 0.5 and less than or equal to 0.8, it revealed a significant correlation; moreover, when it was greater than 0.8, it revealed a high correlation. To intuitively evaluate the contribution of four single herbs, 9 plasma components were successfully screened from the detected components based on the values of the Pearson correlation coefficient ($|r| > 0.5$) as shown in Table 4. The results also provided strong evidence that GR was the principal herb in the formula, whereas PR, PO, and AT served as adjuvant herbs.

4. Discussion

Traditional Chinese medicines (TCM) are frequently used in the treatment of diseases, which reflects the characteristics of multicomponents and multitargets of TCM. At the same



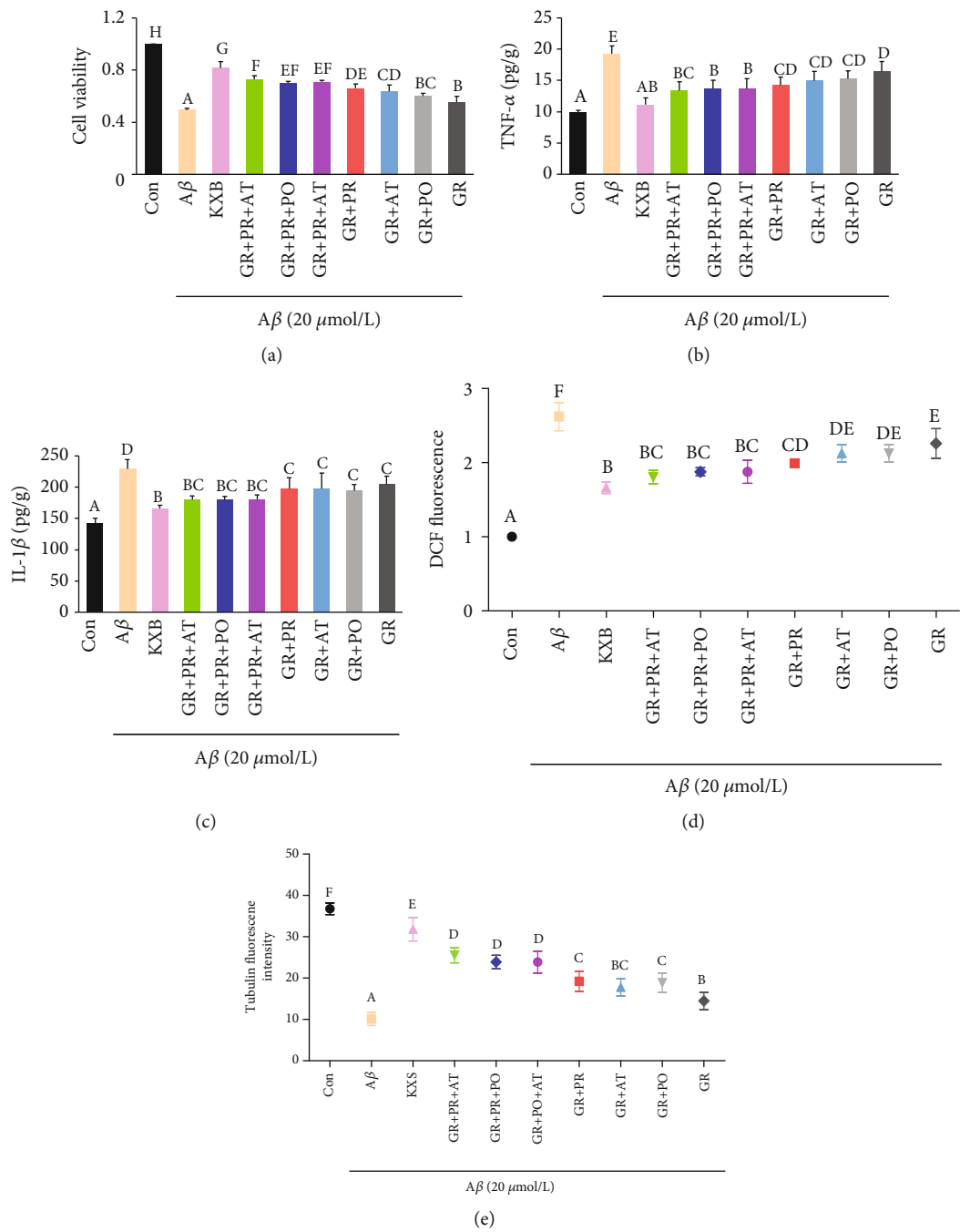


FIGURE 5: Continued.

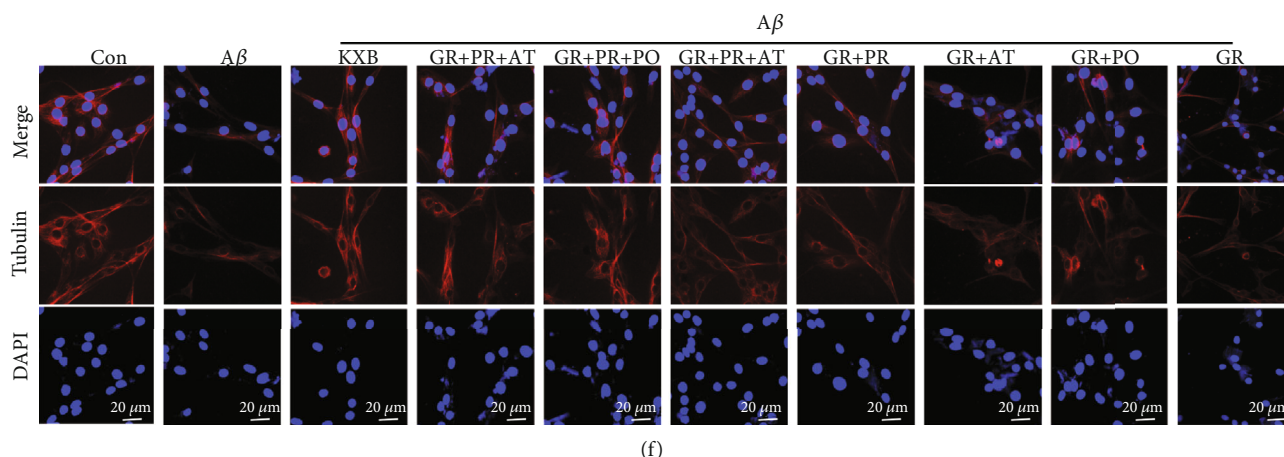


FIGURE 5: *In vitro* therapeutic efficiency of KXS on $A\beta_{25-35}$ -induced PC12 cells. KXS significantly promoted the cell viability of PC12 compared with those treated with PR and/or PO and/or AT combined with the administering of the GR herb (a). Treatment with KXS resulted in the inhibition of inflammation revealed by an alleviation of $TNF-\alpha$ (b) and $IL-1\beta$ (c). Detection of ROS reduction in PC12 cells treated with KXS (d). Morphological examination of Tubulin in PC12 cells treated with KXS and PR and/or PO and/or AT combined with the administering of the GR herb (e and f). The column diagrams marked by different letters were significant to each other at the 0.05 level according to Duncan's multiple range test. Values were mean $\pm$ SE ($n = 6$), scale bar = 20 μm .

TABLE 1: Identification of the components of Kai-Xin-San in plasma samples of rats.

No.	Formula	t_R (min)	Ion addition	Source	Cal m/z	m/z	Fragment ions (m/z)
1	$C_{10}H_{12}O_5$	3.60	$[M + Na]^+$	KXS, AT	235.0577	235.0579	—
2	$C_{12}H_{14}O_5$	31.14	$[M + Na]^+$	KXS, AT	261.0733	261.0729	—
3	$C_{12}H_{16}O_4$	2.14	$[M + Na]^+$	KXS, PR	247.0941	247.0949	—
4	$C_{12}H_{16}O_4$	31.30	$[M + H]^+$	KXS, AT	225.1121	225.1125	—
5	$C_{12}H_{16}O_5$	24.85	$[M + H]^+$	KXS, PO	241.1071	241.1053	186.9109
6	$C_{12}H_{16}O_5$	40.45	$[M + H]^+$	KXS, AT	241.1071	241.1071	—
7	$C_{16}H_{20}O_9$	38.04	$[M - H]^-$	KXS, GR	355.1035	355.1016	—
8	$C_{21}H_{24}O_{11}$	40.12	$[M + H]^+$	KXS, PO	453.1391	453.1385	—
9	$C_{22}H_{30}O_{14}$	48.39	$[M + Na]^+$	KXS, AT	541.1528	541.1535	—
10	$C_{36}H_{46}O_7$	45.95	$[M + H]^+$	KXS, PO	591.3316	591.3316	—
11	$C_{36}H_{56}O_{13}$	49.49	$[M - H]^-$	KXS, AT	695.3648	695.3621	—
12	$C_{36}H_{60}O_8$	46.54	$[M - H]^-$	KXS	619.4215	619.4288	—
13	$C_{36}H_{62}O_{10}$	46.79	$[M - H]^-$	KXS, GR	653.4270	653.4260	277.1787-257.1526-231.1734
14	$C_{41}H_{70}O_{13}$	41.75	$[M - H]^-$	KXS	769.4744	769.4705	339.1989
15	$C_{41}H_{70}O_{14}$	46.33	$[M + H]^+$	KXS, GR	787.4838	787.4866	—
16	$C_{42}H_{72}O_{14}$	45.98	$[M + Na]^+$	KXS, PO	823.4819	823.4792	662.2742-479.1886-343.1892-240.1016-104.1089
17	$C_{47}H_{80}O_{17}$	46.81	$[M - H]^-$	KXS, GR	915.5323	915.5270	277.2151
18	$C_{48}H_{82}O_{18}$	35.4	$[M + Na]^+$	KXS, GR	969.5393	969.5383	789.4727-349.1107-203.0524
19	$C_{48}H_{82}O_{18}$	42.49	$[M + Na]^+$	KXS, GR	969.5393	969.5374	855.5226-539.2863-447.2406-429.2286
20	$C_{53}H_{90}O_{22}$	41.09	$[M + Na]^+$	KXS, GR	1101.5816	1101.5823	—
21	$C_{54}H_{92}O_{23}$	40.74	$[M - H]^-$	KXS	1107.5956	1107.5956	945.5487-783.4916-325.1803-179.0564
22	$C_{54}H_{92}O_{23}$	40.95	$[M + Na]^+$	KXS, GR	1131.5922	1131.5929	789.4755-365.1054
23	$C_{54}H_{92}O_{23}$	45.99	$[M + H]^+$	KXS	1109.6102	1109.6129	—

TABLE 2: Identification of the components of Kai-Xin-San in brain samples of rats.

No.	Formula	t_R (min)	Ion addition	Source	Cal m/z	m/z	Fragment ions (m/z)
1	$C_{12}H_{16}O_4$	20.60	$[M + H]^+$	KXS	225.1121	225.1149	—
2	$C_{12}H_{16}O_4$	28.18	$[M + Na]^+$	KXS	247.0991	247.1049	202.0811-172.9562-131.0022-116.9332-88.0237-80.0488-56.9698
3	$C_{12}H_{16}O_5$	20.27	$[M + Na]^+$	KXS	263.0890	263.0805	78.9849-60.9592-56.9690
4	$C_{12}H_{16}O_5$	40.76	$[M + H]^+$	KXS, GR	241.0971	241.0931	153.0531-91.0597
5	$C_{16}H_{20}O_9$	37.93	$[M - H]^-$	KXS	355.1035	355.1012	297.1512-183.0114-119.0531
6	$C_{21}H_{24}O_{11}$	9.75	$[M + Na]^+$	KXS	475.1211	475.1203	—
7	$C_{21}H_{24}O_{11}$	40.30	$[M + H]^+$	KXS, GR	453.1391	453.1391	328.2223-187.0355-160.9726-124.9905-76.9988
8	$C_{36}H_{46}O_7$	28.07	$[M + Na]^+$	KXS, GR	613.3136	613.3144	—
9	$C_{36}H_{46}O_7$	45.93	$[M + H]^+$	KXS, GR	591.3316	591.3344	532.2582-408.2573-386.2741-146.9808-86.0980
10	$C_{42}H_{72}O_{13}$	40.26	$[M + Na]^+$	KXS	807.4865	807.4872	—

TABLE 3: The correlation of pharmacological indicators and plasma components.

Source	Formula	AChE	ROS	IL-1 β	Pearson's correlation coefficient (r)			Crossing number
					TNF- α	p-Tau (S199)	p-Tau (S396)	
GR	$C_{16}H_{20}O_9$	-0.767	-0.703	-0.855	-0.723	-0.8	-0.744	0.675
GR	$C_{36}H_{62}O_{10}$	-0.751	-0.679	-0.846	-0.696	-0.783	-0.735	0.644
GR	$C_{47}H_{80}O_{17}$	-0.785	-0.731	-0.864	-0.756	-0.819	-0.754	0.714
GR	$C_{48}H_{82}O_{18}$	-0.832	-0.822	-0.867	-0.865	-0.867	-0.765	0.848
GR	$C_{48}H_{82}O_{18}$	-0.835	-0.85	-0.844	-0.903	-0.87	-0.75	0.9
GR	$C_{53}H_{90}O_{22}$	-0.762	-0.695	-0.852	-0.714	0.795	-0.741	0.665
GR	$C_{54}H_{92}O_{23}$	-0.566	-0.439	-0.711	-0.426	-0.591	-0.605	0.346
GR	$C_{41}H_{70}O_{14}$	-0.695	-0.602	-0.81	-0.607	-0.725	-0.699	0.544
PR	$C_{12}H_{16}O_4$	-0.823	-0.9	-0.698	-0.893	-0.782	-0.738	0.821
PO	$C_{12}H_{16}O_5$	-0.335	-0.412	-0.226	-0.519	-0.49	-0.299	0.524
PO	$C_{21}H_{24}O_{11}$	-0.025	-0.077	0.087	-0.182	-0.222	-0.052	0.164
PO	$C_{36}H_{46}O_7$	-0.03	-0.083	0.081	-0.188	-0.227	-0.056	0.171
PO	$C_{42}H_{72}O_{14}$	-0.598	-0.687	-0.506	-0.781	-0.687	-0.503	0.808
AT	$C_{10}H_{12}O_5$	0.04	-0.119	0.036	-0.21	0.137	0.285	0.509
AT	$C_{12}H_{14}O_5$	-0.152	-0.312	-0.137	-0.409	-0.075	0.093	0.678
AT	$C_{12}H_{16}O_4$	0.507	0.388	0.459	0.335	0.632	0.694	-0.033
AT	$C_{12}H_{16}O_5$	-0.032	-0.193	-0.029	-0.287	0.057	0.214	0.576
AT	$C_{22}H_{30}O_{14}$	-0.299	-0.454	-0.271	-0.553	-0.242	-0.064	0.787
AT	$C_{36}H_{56}O_{13}$	0.166	0.012	0.15	-0.072	0.274	0.404	0.382

TABLE 4: The number of components in four single herbs in AD rat plasma.

Number of cases	GR	PR	PO	AT
All	8	1	4	6
$ r > 0.5$	7	1	1	0

time, it also conforms to the pathological mechanisms of most diseases caused by multiple causes, especially chronic diseases and complex diseases [38]. The selection of com-

pound formulae is based on the integral theory of TCM and follows the rule of drug compatibility and synergism [39]. However, it is still a great challenge to find out the direct targets and explore the mechanisms of multitarget and multichannel actions of the TCM compound. Trying to explore the possible working mechanism of the formula with systems biology approaches, we used KXS as an example since it is certified to have a curative effect in treating AD.

In our study, it was found that, in the AD rat model treated with KXS consisting of four single herbs, there was a significantly shortened escape latency and swimming distance in the Morris water maze to improve cognitive

impairment. In the KXS treatment group, AChE, a characteristic hydrolase of ACh, was significantly reduced and could further regulate the level of ACh. These results, together with the immunofluorescence of NeuN, suggested that KXS play a crucial role in the anti-AD therapeutic effect. Satisfactorily, KXS did not reveal any obvious side effect in AD rats, as shown by weight change, consistent with the meeting of the clinical medication safety committee. In order to explore the possible working mechanisms, a study of the related protein signaling pathways was particularly necessary [40]. Plenty of researches have confirmed that Tau was the vital pathogenic factor and important target for drug development [41, 42]. Interestingly, KXS exerted the best pharmaceutical effect in regulating Tau signaling pathways, and among the four single herbs of KXS, GR has also been regarded as the primary single herb aiming at Tau hyperphosphorylation. In AD rat research, KXS was discovered to be able to upregulate p-PI3K, p-Akt, and p-GSK-3 β and further lead to the downregulation of p-Tau (S199 and S396) in the hippocampus of AD rats. In the cellular level, it was found that KXS could modulate Tau hyperphosphorylation by the PI3K/Akt signal pathway, while Tau, as the microtubule-associated protein, could play a role on the cytoskeleton of nerve cells through maintaining the stability of microtubules. The therapeutic effect of GR was the most effective among the four single herbs, and the effects of GR coupled with PR, PO, or AT were further enhanced to a certain degree. Furthermore, Tau was not only regulated by phosphorylation but also by truncation, glycosylation, acetylation, and other posttranslational modifications [43]. In the pathological process of AD, the Tau protein could be truncated and could induce the apoptosis of neurons. The enzyme involved in abnormal Tau truncation was mainly caspase-3, which also indicated that KXS might be related to the regulation of apoptosis [44, 45].

Maintenance of the normal physiological status required a balance of the production and cleanup of free radicals [46]. KXS could regulate oxidative stress both *in vivo* and *in vitro*; specifically, intracellular ROS could be reduced by GR, PR, PO, and AT, particularly when the four herbs were adopted in combination. Furthermore, it was known that, in the synergistic and compatible rule of designing the formula in TCM, some herbs should play an adjuvant role in facilitating the delivery of the sovereign drug to the disease site *in vivo* [47]. Then, formula dismantling *in vitro* indicated that ginseng was the principal herb of the formula, whereas the other three herbs served adjuvant roles to achieve the best antioxidant effect of the whole formula. Meanwhile, neuroinflammation was also an important inducement of pathological changes in AD and interacted with oxidative stress [48, 49]. In the KXS treatment group, TNF- α and IL-1 β , the critical proinflammatory factors, were significantly downregulated both *in vivo* and *in vitro*. It was found that GR showed the strongest antineuroinflammation effect in four single herbs and the effects of GR coupled with PR, PO, or AT were further enhanced to a certain degree. These results suggested that PR, PO, and AT might assist GR in the KXS formula through mediating neuroinflammation. Several studies suggested that a vicious circle between

oxidative stress and neuroinflammation was a significant pathogenic feature in AD, and the circle might aggravate the A β -induced apoptosis and the hyperphosphorylation of Tau [50, 51]. In the AD model group, apoptosis-related proteins had high expression, whereas the KXS formula significantly inhibited the expressions of Bax, bcl-2, and cleaved-caspase-3. This phenomenon not only suggested that KXS might regulate oxidative stress and neuroinflammation-triggered apoptosis but also indicated that the formulae might have therapeutic effects through mechanisms beyond each herb.

The application of the KXS formula has been retrospectively to the Tang dynasty, and the explanation of the basis of its potential pharmacodynamic material could also be achieved by modern analytical techniques [52]. At the small molecule level, the LC-Q-TOF-MS method with high resolution and sensitivity was established to analyze the transitional components of KXS *in vivo*. The mass spectra analysis showed that 23 components were firstly detected in AD rats' plasma, including 8 from GR, 1 from PR, 4 from PO, 6 from AT, and 4 that only existed in KXS; meanwhile, 10 constituents were firstly found in AD rats' brain, including 4 from GR and 6 that were only discovered in KXS. Through semiquantitative analysis, it is suggested that the peak areas of some components in the KXS group were larger than that in the GR group. Moreover, it was found that the herbs' combination could facilitate the transportation of GR into blood and through the blood-brain barrier. The results not only verified the principal status of ginseng but also indicated that the formula might have better anti-AD effects than each herb in the formula. These phenomena reflected the synergistic effect and the formula compatibility rationality of KXS to some degree. In order to further explore the principles of formulating prescription, assessment of the coefficient value by the Pearson correlation analysis was applied to explain the correlation between plasma components and pharmacological function. The correlation analysis results (Tables 3 and 4) suggested that the components in GR showed the strongest correlation with anti-AD effects based on the values of the correlation coefficient ($|r| > 0.5$), as well as reflected the action of the multicomponent, multichannel, and multitarget characteristics of traditional Chinese medicine.

In conclusion, the data not only suggested that KXS could ameliorate AD by regulating the cholinergic system, Tau hyperphosphorylation, oxidative stress, neuroinflammation, and apoptosis in the AD model but also indicated that GR is the principal herb of the formula, whereas PR, PO, and AT acted as adjuvant herbs. The results acquired by systems biology methods at the animal, cellular, and molecular levels suggested that the dissection of the mechanism of the clinically well-used formula could provide a novel way in exploring the value of TCM.

Data Availability

The data used to support the findings of this study are included within the article and the supplementary information files.

Conflicts of Interest

The authors announce no conflict of interest to the current research.

Authors' Contributions

Sirui Guo and Jiahong Wang contributed equally to the work.

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Supplementary Materials

HPLC-QTOF MS was applied to establish a method for semiquantitative analysis of components into plasma and brain of AD rats. Data processing methods, such as the multiple product ion filtering technique and the neutral loss filtering technique, were adopted, and the components were assigned by formula disassembly. The peak areas of components after oral administration of KXS are listed in Table S1 and Table S2. (*Supplementary Materials*)

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